

ANIRBAN DEY

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EDUCATIONAL QUALIFICATION

University of Maryland, College Park, MD
Master of Engineering, Software Engineering

May'25

SKILLS

Languages: Java, JavaScript, SQL, HTML, CSS, TypeScript

Frameworks & Libraries: Spring Boot, ReactJS, Electron, Hibernate, JUnit, Mockito, Express.js, Socket.IO, TailwindCSS

Databases: Oracle 19c, PostgreSQL, IBM DB2, Microsoft SQL Server, MySQL, Cassandra, MongoDB, Neon, Supabase, MariaDB

Tools: Node.js, RHEL, Terraform, Vite, Jenkins, Kafka, Redis, AppDynamics, Nexus IQ, Checkmarx, Splunk, Docker, Kubernetes (OpenShift), Red Hat Quay, ServiceNow, Vercel, IBM API Connect, IBM DataPower, Stripe, Clerk, Async Profiler, Nodemailer

Cloud & Infra: AWS(ECS, VPC, CDK, MSK, RDS, CloudWatch, CloudFront, S3, Lambda, DynamoDB), IBM WebSphere, Tomcat

WORK EXPERIENCE

INFOSYS | CVS Aetna

Bangalore, India

TECHNOLOGY ANALYST

Sep'22-Aug'23

- Spearheaded a **\$1M modernization** effort to upgrade **25+ legacy applications (~300K LOC)** from **Java 8 to Java 17**, resulting in a **90% reduction in disk usage** and up to **40% improvement in application performance**.
- Managed and mentored a **16-member development team**, delivering **high-impact solutions** on schedule and securing a **5-year client contract renewal** through strategic planning, quality execution, stakeholder alignment.
- Remediated **security vulnerabilities** flagged by **Nexus IQ** and **Checkmarx**, enhancing **application resilience** and reducing potential threats by **25%** through **proactive code analysis** and mitigation strategies.
- Integrated **Redis** as a **low-latency caching layer** for frequently accessed APIs, reducing database **query load by 70%** and cutting **average response times from 450ms to under 100ms**—accelerating **claims processing** for over **1 million CVS Aetna users**.
- Leveraged **Async Profiler** for **low-overhead CPU and memory profiling** in **Spring Boot microservices**; analyzed **flame graphs** to identify **thread contention**, **memory leaks** and **GC bottlenecks**, then optimized **thread pool sizes**, refined **garbage collection strategies**, and reconfigured **JVM parameters**, achieving a **30% increase in throughput**

SENIOR SYSTEMS ENGINEER

Feb'21- Sep'22

- Optimized and stabilized **10+ Java/J2EE applications** in a **high-traffic production environment**, reducing **system crashes by 40%** and ensuring uninterrupted service for over **1 million active users**.
- Engineered **scalable WebRTC infrastructure** supporting **10,000+ real-time peer sessions** with **<200ms latency**, integrating custom **STUN/TURN servers**, **SFU-based media routing**, and **adaptive bitrate control** for network efficiency.
- Engineered **OpenShift pods and containers** for **high-availability Java applications**, implementing **health checks**, **auto-scaling**, & **resource quotas** to ensure **optimal performance & fault tolerance** in secure production environments.
- Configured **Splunk** and **AppDynamics** dashboards for **real-time monitoring** & cutting **issue resolution time** in half.

SYSTEMS ENGINEER

Nov'19-Feb'21

- Conducted **comprehensive unit testing** of applications using **JUnit** and **Mockito**, ensuring **robustness & reliability of critical features** through **rigorous test case design & execution**, leading to **30% decrease in post-release defects**.
- Resolved **batch job failures** & optimized **scheduling logic**, cutting **downtime by 40%** & boosting **system availability**.
- Automated **Jenkins CI/CD pipelines** cutting **deployment errors** and **manual work** while accelerating **release cycles**.

PROJECT

[Rekord- AI Driven Realtime Screen Recording & Video Sharing Platform](#)

- **Built a cross-platform desktop app** using **Electron**, **React**, and **Vite**, featuring a **multi-window architecture** (main, studio, webcam) and **1080p/60fps capture** with real-time preview via **IPC Renderer**.
- **Designed and deployed a scalable backend pipeline** using **Express.js**, **AWS S3**, and **CloudFront**, supporting **1,000+ concurrent 1080p streams** with **sub-100ms video processing latency** via **WebSocket** communication.
- Integrated **AI features** using **OpenAI Whisper API** for transcription and Q&A, reducing **manual content processing time by 80%** and enabling **searchable video playback**.
- Achieved **99.9% uptime** through global **CDN distribution** via **CloudFront** and **real-time streaming infrastructure**.

[MediSync- Healthcare Microservices System](#)

- Built scalable Spring Boot microservices for **10,000+ concurrent users** by designing a **gRPC+Kafka communication layer** with **API Gateway** for **routing, authentication, and rate limiting**—cutting **response times by 40%** and **messaging latency by 60%** through **asynchronous, fault-tolerant orchestration**.
- Deployed **containerized workloads** on **AWS ECS** within a **custom VPC** using **AWS CDK**, integrating services like **MSK**, **RDS**, and **CloudWatch**; optimized deployments reduced **provisioning time by 75%** and cut **infrastructure costs by 30%**.